ANNEX TECHNOLOGIES · DATA ENGINEER CASE STUDY

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ABC Phones installment credit portfolio

From raw files to a monitored pipeline and a dashboard, and what the data says about credit risk and customer satisfaction.

71,456
loan records across 5 quarterly snapshots

19,565
financed loans in the December 2025 snapshot

48
automated data quality checks on every run

PART 1A · PROFILING AND CLEANING

The raw files had seven problems, all handled and logged

Problem found	Size	What we did
Loan Id is #N/A on most demographic rows	76% of birth-date rows	Dropped, loan cannot be identified
Credit RETURN_DATE is corrupted	All 71,456 rows	Column dropped
Last payment date falls after the snapshot	31.5%, 22,473 rows	Flagged, kept out of features
File named 30-03 holds 31 March data	1 of 5 snapshots	Date read from the data
CUSTOMER_AGE is days since sale, not age	All rows	Age rebuilt from date of birth
Loan term not numeric, such as STDHIGH	765 sales rows	Raw value kept and flagged
Same loan answered the NPS survey twice or more	533 loans	Latest response flagged

No credit row was lost: 71,456 in, 71,456 out. Of the 20,742 loans in the latest snapshot, 11,175 have a valid age band, 10,609 have income and 3,532 have an NPS response.

PART 1C · ETL DESIGN

One pipeline from raw files to the dashboard

Source files
5 credit CSVs, sales and NPS workbooks. Never edited, archived in Cloud Storage.

→

Clean and mask
Python and pandas. Masking key from Secret Manager.

→

Staging
BigQuery abc_staging, 5 masked tables.

→

Quality gate
48 checks before anything is built.

→

Features, marts
SQL builds abc_marts: 1 fact table, 3 metric tables.

→

Dashboard
Looker Studio, 3 pages.

Ingestion

Full reload on every run, safe at 71,456 rows. A daily check looks for new files. Identical files are skipped and a corrected snapshot replaces the old one.

If a critical check fails

The run stops and an ALERT is logged. The marts keep the last good data.

Storage and access

Fact table partitioned by snapshot date, clustered by loan. Analysts use Looker Studio or SQL on abc_marts.

Production option, not deployed: Cloud Scheduler starts a Cloud Run job that runs the same script. Full diagram: pipeline_design/architecture.png

PART 1B · FEATURES

Four features, built as at each snapshot and masked at the door

Age band

`age_band`

Age on each snapshot date, from date of birth, which is then discarded. Bands 18-25, 26-35, 36-45, 46-55, 56+.

Income band

`avg_monthly_income_band`

Amount received divided by Duration, a 1 to 24 month statement window. Eight bands, Below 5,000 to 150,000+.

Days past due

`days_past_due`

Zero when arrears are not positive. The reported value is kept beside it for comparison.

Risk category

`risk_category`

Seven rules, first match wins. Critical: written off, returned or 61+ days late. High: first-payment default or 31-60 days. Medium: 8-30 days or earlier lateness. Low: current or paid off.

Masked at the door

Every ID is an HMAC-SHA256 hash. The key is in Secret Manager. A guard blocks any raw ID, email or phone number.

Flag, never overwrite

Quality flags sit in dq_ columns beside the source values. History features use earlier snapshots only.

Simple stack

Python for messy parsing, SQL for features. Airflow or dbt would be overkill for three source files.

PART 1C · ERRORS AND RECOVERY

Every failure has a defined outcome, and re-running is safe

What goes wrong	What happens	How we know
A source file is missing	Cleaning stops with a clear message	ERROR log, failed step
A snapshot has not arrived	Gate stops the run, last good marts keep serving	ALERT line, dq_results
A file cannot be parsed	Crash before any write, retried twice, then stop	ERROR log, failed step
A critical check fails	Run stops and is not retried	Exit code 2, ALERT line
Personal data would reach a table	Write refused, marts scanned again	ERROR log, exit code
A cloud call fails	Step retried, no double load	Warning per attempt
The pipeline runs twice	Nothing changes, identical files are skipped	load_audit checksums

Why a retry is safe

Loads replace whole tables, so a retry cannot double-load. The marts are rebuilt only after the quality gate passes, so the dashboard never reads half-built data.

PART 2 · DATA QUALITY FRAMEWORK

48 automated checks caught a real source defect

Check family	Count	Catches
Freshness	4	Missing or shrinking snapshots
Uniqueness	5	Duplicate loans and responses
Referential integrity	7	Orphan loans across tables
Null thresholds	13	Missing keys, dates, amounts
Range	7	Impossible values and dates
Schema contract	5	Column or type changes
Personal data scan	7	Raw IDs, emails, phones

`scripts/quality_checks.py` · `sql/04_dq_checks/`

Alerting. Critical: engineer at once, business owner next morning, engineering lead if still open after 4 hours. Warning: email and weekly digest. Info: recorded.

Cadence. Real time, on every load: files, row counts, keys, personal data. Daily: all 48 checks. Weekly: trend and drift review.

REAL ANOMALY, RUN OF 5 OCT 2026

31% of rows have a last payment date after the snapshot date

The extract stamps this field when it is exported, so older snapshots contain future information. It is flagged and kept out of features and machine learning.

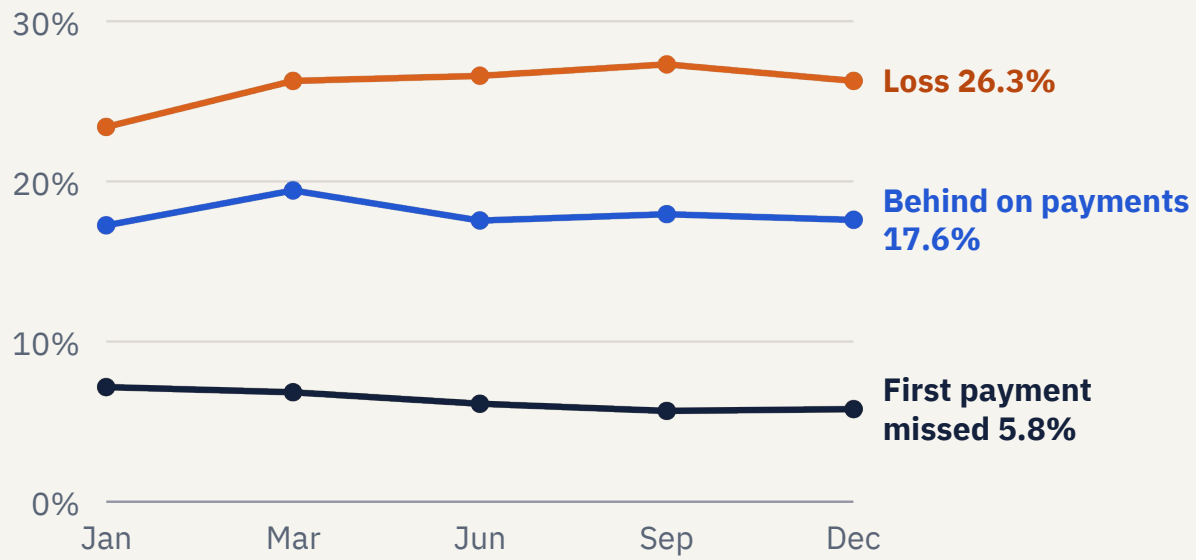
```
WARNING [warning]
range.payment_date_after_snapshot_share: observed
0.3145 vs threshold 0
43 checks: 42 passed, 1 failed (0 critical)
```

Log from the first full run, before the 5 schema checks were added.

PART 3A · PORTFOLIO HEALTH, FIVE SNAPSHOTS OF 2025

Delinquency is steady, loss stays high, first-payment default is improving

Risk rates at each snapshot (lower is better)



Money collected

94.2%

94.6% in Jan, low 93.1% in Jun. Recovering.

Higher is better

Customers retained

95.8%

93.6% in Mar, 95.8% now. Improving.

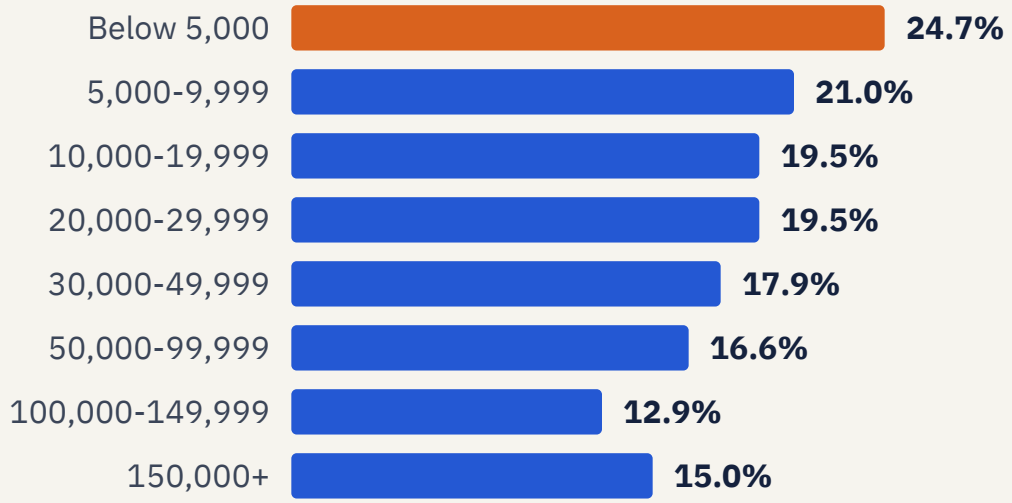
Higher is better

Snapshots: 1 Jan, 31 Mar, 30 Jun, 30 Sep, 30 Dec 2025. Delinquency and collection use performing loans. Loss is lost loans over all financed loans. Retention starts in March because it compares each snapshot with the one before.

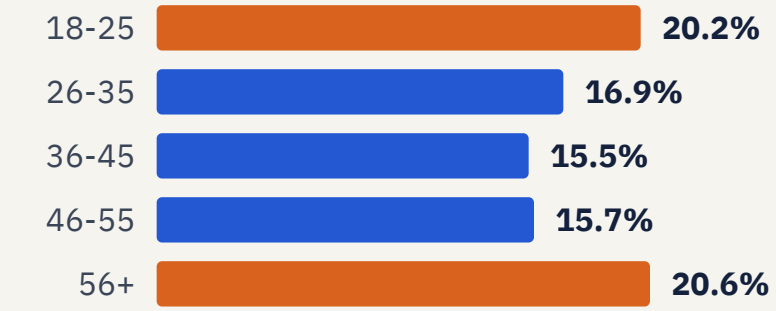
PART 3A · WHO FALLS BEHIND

Customers on lower incomes fall behind more often

Behind on payments, by monthly income (average 17.3%)



Behind on payments, by age



Income shows the clearer pattern. The youngest and oldest groups also fall behind more than ages 26 to 55.

The segment that stands out: income below 5,000 at 24.7% against an average of 17.3%. Loans sold since 1 Jan 2025, December 2025 snapshot. Delinquency falls steadily as income rises (trend test $p < 0.0001$). Below 5,000 has only 170 performing loans and 56+ only 73, so those two bars are the least certain.

PART 3B · CREDIT AND CUSTOMER SATISFACTION

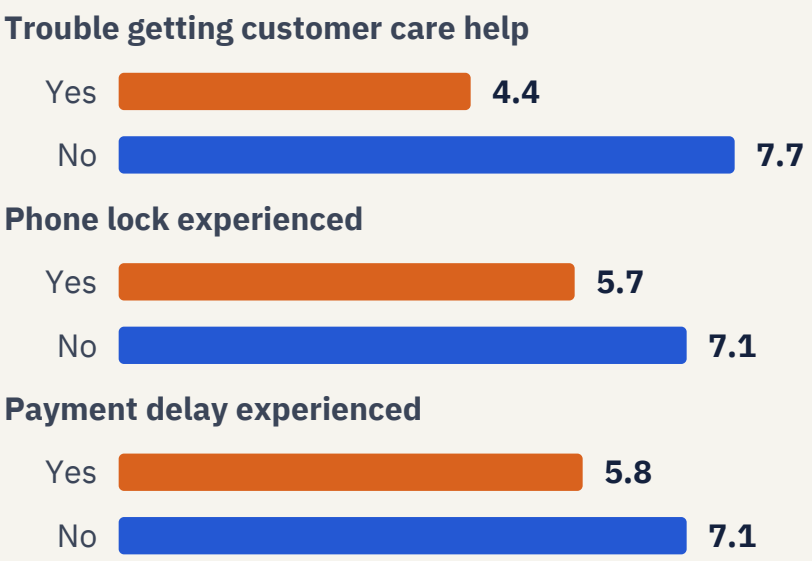
Customers who fall behind score us lower

Average score (0 to 10), by how late



There is no credit score in the data, so days late stands in for it. NPS: +6.5 up to date, -39 at 91+ days late.

Average score, by experience



RECOMMENDATION

Contact customers in the first 7 days of arrears, make customer care easy to reach, and offer a simple payment plan. Test it against a control group.

Collections and satisfaction are in tension: customers who experienced a phone lock score 5.7 against 7.1. Trouble getting help rises from 32% when up to date to 45% at 1-7 days late.

PART 3C · DATA GAPS AND NEXT STEPS

What the data lacks, and what to improve first

What is missing

- A working Loan Id on the demographic sheets
- A customer id, so one person’s loans cannot be linked
- Employment type and location
- Payment-level history
- Write-off date and amount
- A credit score

What is inconsistent or unclear

- Mixed date formats and #N/A tokens
- Last payment date stamped at extract time
- Write Off means 91+ days past due
- PAR 30 mostly holds write-offs
- What Duration means (a statement window)
- Loan terms such as STDHIGH

Three improvements

1. **Stable keys.** One customer id and one loan id, in every file.
2. **History with dates.** Status and payments with effective dates.
3. **A data contract.** Validate each file at the source before delivery.

NEXT: MACHINE LEARNING

Predict which loans reach 91+ days or Critical by the next snapshot, using features as at each snapshot only and a split by time. With only 4 snapshot transitions, results will be indicative.